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THE STELLAR DISTRIBUTION AROUND A BLACK HOLE: NUMERICAL INTEGRATION OF THE FOKKER-PLANCK EQUATION

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ABSTRACT

We determine the steady-state stellar distribution around a central black hole in a star cluster by means of a direct numerical integration of the Fokker-Planck equation in energy–angular momentum space. The loss cone in phase space resulting from tidal destruction of stars is treated by means of a detailed boundary layer analysis. The process of stellar destruction by direct physical collisions is treated by use of the physical collision cross section.

We present the two-dimensional steady-state distribution function, the density and velocity dispersion profiles of the stellar distribution, and stellar consumption rates for black holes in globular cluster environments. The distribution function we obtain is in reasonable agreement with that resulting from the Monte Carlo simulations of Shapiro and Marchant; our loss rate is larger than theirs by a factor of 2.

Subject headings: black holes — clusters: globular — stars: stellar dynamics

I. INTRODUCTION

The determination of the steady-state stellar distribution in the vicinity of a massive black hole residing in the core of a stellar system has recently received a good deal of attention (Peebles 1972; Bahcall and Wolf 1976, 1977; Frank and Rees 1976; Lightman and Shapiro 1977; Young 1977*a*, *b*; Ozernoy and Dokuchaev 1977; Ipser 1978*a*, *b*; Shapiro and Marchant 1978). These investigations have been given impetus by the suggestions that black holes may power X-ray sources in globular clusters (Bahcall and Ostriker 1975; Silk and Arons 1975), active galactic nuclei, and quasars. The mathematical problem which these authors considered and to which we address ourselves in this paper is as follows.

a) Statement of Problem

The time-dependent stellar distribution around a black hole can be characterized by a distribution function $f(\mathbf{r}, \mathbf{v}, t)$ which must satisfy the Boltzman equation:

$$\left(\frac{\partial f}{\partial t}\right)_v \equiv \frac{\partial f}{\partial t} + \mathbf{v}\cdot\nabla f + \nabla\phi\cdot\nabla_v f = \left(\frac{\partial f}{\partial t}\right)_c + \left(\frac{\partial f}{\partial t}\right)_p , \tag{1}$$

where ϕ is the gravitational potential,[1] the subscript c refers to binary encounters, and the subscript p refers to actual physical collisions. We neglect large-angle binary scatterings so that $(\partial/\partial t)_c$ reduces to the Fokker-Planck collision operator (Rosenbluth, MacDonald, and Judd 1957). We restrict ourselves throughout this paper to stars of only a single mass m. A simplification which has commonly been applied to this problem is the neglect of the self-potential of the stellar distribution. In this approximation the gravitational potential is just that due to the black hole of mass M and is taken to be Keplerian in the region of interest: $r_t < r < r_M$, where r_t is the tidal break-up radius of the stars under consideration in the field of the black hole and r_M is the distance out to which the gravitational force exerted by the black hole dominates that of the star cluster. The usual boundary conditions adopted are that f vanishes at r_t and that f smoothly joins the core distribution of the stellar system at $r \sim r_M$.

For realistic stellar systems the problem has a natural small parameter, $\lambda(E) \equiv P/T_r$, where P is the Keplerian orbital period of a star bound to the black hole with binding energy E and T_r is the relaxation time in the core of the stellar system. For globular cluster stars with $E \gtrsim GM/r_M$, $\lambda(E) \lesssim 10^{-4}$. Since the ratio of the collision term (right-hand side of eq. [1]) to the Vlasov term (left-hand side) is of order λ, a perturbation expansion of the Boltzmann equation in λ is appropriate. Take

$$f = f_0 + f_1 , \tag{2}$$

where

$$(\mathbf{v}\cdot\nabla + \nabla\phi\cdot\nabla_v) f_0 = 0 , \tag{3a}$$

$$\frac{\partial f_0}{\partial t} + (\mathbf{v}\cdot\nabla + \nabla\phi\cdot\nabla_v) f_1 = \left(\frac{\partial f_0}{\partial t}\right)_c + \left(\frac{\partial f_0}{\partial t}\right)_p . \tag{3b}$$

[1] We adopt the opposite of the usual sign convention for ϕ; thus the equation of motion is $d\mathbf{v}/dt = \nabla\phi$.

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By Jeans's theorem the solution to (3a) has the form $f_0 = f_0(E, J, t)$, where E is the *binding* energy per unit mass (negative of the usual energy) and J is the angular momentum per unit mass. The dependence of f_0 on t is slow, so that $\partial f_0/\partial t$ is of first order in λ. In terms of E and J, the velocity space volume element can be written:

$$d^3v = 4\pi \frac{JdJdE}{r^2 v_r}, \tag{4}$$

where $v_r = [2(\phi - E) - (J/r)^2]^{1/2}$ is the radial velocity of a star of energy E and angular momentum J at radius r. [To take into account the fact that there are two radial velocities of opposite sign which correspond to the same set of values (E, J, r), an extra factor of 2 is included in d^3v.] To find the slow evolution of f_0 with time, we make use of equation (3b), which has a single-valued solution for f_1 if and only if its time average along every orbit in phase space vanishes. We first transform (3b) to E, J, r, t coordinates and divide by v_r, obtaining

$$\frac{\partial f_1(E, J, r)}{\partial r} + \frac{1}{v_r}\frac{\partial f_0}{\partial t} = \frac{1}{v_r}\left[\left(\frac{\partial f_0}{\partial t}\right)_c + \left(\frac{\partial f_0}{\partial t}\right)_p\right]. \tag{5}$$

For fixed E and J, the radial coordinate r of a star is restricted to lie between the turning points $r_-(E, J)$ and $r_+(E, J)$ which are given by equation (18). At each point r in this range, E and J determine the motion of a star up to the sign of the radial velocity. Thus f_1 may be written

$$f_1(E, J, r) = \tfrac{1}{2}[f_1^{\text{out}}(E, J, r) + f_1^{\text{in}}(E, J, r)], \tag{6}$$

where the superscripts indicate the direction of radial motion. (The factor of $\frac{1}{2}$ is included because of the extra factor of 2 in d^3v.) At $r_\pm$, where v_r vanishes, we have the boundary condition

$$f_1^{\text{out}}(r_\pm) = f_1^{\text{in}}(r_\pm). \tag{7}$$

Let the "orbital integral" of a function $F(r, v_r)$ be defined by

$$\oint drF(r, v_r) \equiv \int_{r_-}^{r_+} drF(r, v_r) + \int_{r_+}^{r_-} drF(r, -v_r). \tag{8}$$

The orbital integral of equation (5) reduces to

$$\frac{\partial f_0}{\partial t} = P^{-1}\oint \frac{dr}{v_r}\left[\left(\frac{\partial f_0}{\partial t}\right)_c + \left(\frac{\partial f_0}{\partial t}\right)_p\right] \tag{9}$$

when we apply the boundary condition (7). Here $P = \oint dr/v_r = 2^{-1/2}\pi GME^{-3/2}$ is the Keplerian period for a star of energy E in the field of a black hole of mass M. Equation (9) is the orbit-averaged Fokker-Planck equation in (E, J)-space as given, for example, by Lightman and Shapiro (1977). The steady state condition for f_0 may be written:

$$\oint \frac{dr}{v_r}\left[\left(\frac{\partial f_0}{\partial t}\right)_c + \left(\frac{\partial f_0}{\partial t}\right)_p\right] = 0. \tag{10}$$

Stars which come within the tidal breakup sphere of radius r_t will be torn apart by the tidal force of the massive black hole and will be destroyed. (Actually, if the pericenter of the orbit of a star is comparable with r_t, several passes are probably needed to completely tear the star apart; we neglect this complication in our treatment.) Thus, we have the boundary condition at r_t,

$$f(r_t, \boldsymbol{v}) = 0. \tag{11}$$

Transforming this condition to E and J coordinates, we see that f_0 must vanish for those orbits whose pericenters lie closer than r_t to the black hole. This delineates a region in (E, J)-space such that stars within this region will be destroyed by the black hole if they pursue their orbits unaffected by collisions. We must have $f_0(E, J) = 0$ throughout this region which is defined by

$$0 \le J^2 \le J_{\min}^2(E) \equiv \frac{(GM)^2}{E_t}\left[1 - \frac{1}{2}\frac{E}{E_t}\right], \tag{12}$$

where $E_t = GM/2r_t$. This region has commonly been labeled the "loss cone" by analogy with a similar domain in the phase space of a plasma confined in a mirror machine. At any space point all stars with velocity vectors directed within a hyperboloid of one sheet whose axis is the radial direction will subsequently pass within a distance r_t of the black hole. The equation of this hyperboloid in velocity space, which follows from equation (12), is

$$v^2_{t,\min} = \frac{2r_t}{r + r_t}\phi + \frac{r_t^2}{r^2 - r_t^2}v_r^2, \tag{13}$$

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where $v_{t,\min} \equiv J_{\min}/r$. For $r \gg r_t$ the portion of the hyperboloid which lies interior to a sphere of radius $v_{\max} = (2\phi)^{1/2}$ (the maximum velocity of a *bound* star) is essentially cylindrical with radius $v_{t,\min} \approx (r_t/r)^{1/2} v_{\max}$. Hence for $r \gg r_t$ the "loss cone" is, more properly, a "loss cylinder" in velocity space. We shall nevertheless use the term "loss cone" to denote the corresponding regions in velocity space and (E, J)-space containing stars with pericenters of less than r_t. The loss cone is of importance because even stars very far from the black hole may be scattered into it and lost along highly elongated orbits.

The loss cone introduces a nonuniformity into the λ expansion. The range of binding energies for stars about the black hole is large since $E_t \gg \langle v^2 \rangle$, the one-dimensional velocity dispersion in the cluster core. Thus, for small binding energies ($E \sim \langle v^2 \rangle$), $J_{\min}{}^2 \ll J_{\max}{}^2 \equiv G^2 M^2/2E$ ($J_{\max}$ is the angular momentum of a circular orbit of energy E), and the λ expansion leads to very sharp gradients in the angular momentum direction in (E, J)-space. Small angle collisions are then of importance, and the collision term is amplified by the inverse square of the opening angle of the loss cone, $\theta = J_{\min}/J_{\max} = (E/E_t)^{1/2}$. Thus we expect the λ expansion to break down in the loss cone region when $\theta^2 \sim \lambda$, and this occurs for energies in the range of interest. Consequently, an asymptotic analysis different from the λ expansion must be made in this region (see § III*b*).

b) Previous Investigations

Most investigations of the distribution of stars around a black hole in a cluster have involved the solution of the problem outlined in § I*a*. These investigations made use of the λ expansion in some form in addition to invoking a variety of simplifying assumptions. In an early analysis, Peebles (1972) introduced the ansatz that f is isotropic in velocity space (independent of J) and has a power law energy dependence, $f \propto E^p$. Using heuristic dimensional arguments based on the assumption that the flux of stars through any energy must be independent of energy, Peebles concluded that $p = \frac{3}{4}$. Bahcall and Wolf (1976) relaxed Peebles's restrictions, taking f to be a general function of E but still independent of J. This form is too special to be a steady-state solution of equation (9). However, if one ignores the loss cone region, as Bahcall and Wolf did in their first paper, the J dependence would be expected to be weak. To determine f, they required that it satisfy the reduced equation obtained by multiplying equation (9) by J and integrating over J to obtain

$$\frac{\partial f(E, t)}{\partial t} = \frac{2}{J_{\max}{}^2} \int_0^{J_{\max}} JdJ \left[\frac{1}{P} \oint \frac{dr}{v_r} \left(\frac{\partial f}{\partial t} \right)_c \right], \tag{14}$$

where $(\partial f/\partial t)_p$ has been neglected. This equation uniquely determines f as a function of E, but of course its solution need not satisfy the more general equation (9). However, if one replaces the exact boundary condition $f(r_t) = 0$ by $f(E_t) = 0$, the J-dependent solution of equation (9) is reasonably close to that of equation (14).

Bahcall and Wolf (1976) found that their numerical solution, which satisfies the boundary conditions that f becomes Maxwellian in the small binding-energy limit and that f vanishes at E_t, is reasonably approximated by $f \propto E^{1/4}$ away from these boundaries. This analytic distribution has a vanishing flux of stars in E-space while Peebles's (1972) solution has an infinite flux away from the black hole.

In another paper discussing the stellar distribution around a black hole Frank and Rees (1976) emphasized the importance of loss cone effects which were not included in the studies of Peebles (1972) and Bahcall and Wolf (1976). Frank and Rees pointed out that, in a quasi-equilibrium state, the dominant mechanism by which stars are consumed by the black hole is the diffusion of stars into very elongated orbits with pericenters within the tidal breakup sphere. They presented an analytic estimate of the loss rate which considerably exceeds the numerical flux calculated by Bahcall and Wolf (1976). They also pointed out the importance of physical collisions which truncate the distribution at energies smaller than E_t.

Lightman and Shapiro (1977) considered loss cone effects using a two-dimensional Fokker-Planck equation in (E, J)-space. They found that the process of angular momentum diffusion into and out of loss cone orbits dominates the solution at energies larger than a critical energy $E_{\rm crit}$, at which the root mean square diffusion step in J per orbital period is equal to the width $J_{\min}$ of the loss cone. (This is the region of nonuniformity of the λ expansion mentioned above.) They estimated loss rates under the assumption that the energy profile of the distribution function can be adequately represented by a power law of constant index. However, they did not attempt to obtain an accurate solution to the two-dimensional Fokker-Planck equation.

Bahcall and Wolf (1977) in a second paper included the loss cone heuristically, following Lightman and Shapiro (1977). In order to semi-independently estimate the importance of loss cone effects, while still employing an angular momentum independent f, Bahcall and Wolf (1977) computed a numerical steady state solution of an equation of the form,

$$\frac{\partial f(E, t)}{\partial t} = F(E, t) - \rho_{\rm lc}(E, t), \tag{15}$$

where F represents the right-hand side of equation (14) and the loss term $\rho_{\rm lc}$ is essentially that given by Lightman and Shapiro (1977). Bahcall and Wolf (1977) find that the addition of this loss term to their basic energy diffusion equation has very little effect on the slope, $d \ln f/d \ln E$, of the steady-state solution $f(E)$. However, it increases the loss rate significantly, as surmised by Frank and Rees (1976) and Lightman and Shapiro (1977).

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Ipser (1978*a*, *b*) treats the Boltzman equation in (position, velocity space)-coordinates and thus does not explicitly invoke the λ expansion. He instead seeks a solution $f(r, v)$ to the *local* Fokker-Planck equation $(\partial f/\partial t)_c = 0$, with the stipulation that f be approximately a function of E and J. Ipser's solution to the Boltzmann equation, $(\partial f/\partial t)_V = (\partial f/\partial t)_c$, may thus be written $f(r, v) = g(E, J) + \delta g(E, J, r)$ with $\delta g/g \ll 1$. It is clear that the term $(\partial \delta g/\partial t)_V$ [of order $\lambda^{-1}(\partial \delta g/\partial t)_c$], which Ipser neglects, is at least as large as the local Fokker-Planck collision term $(\partial f/\partial t)_c$ unless $\delta g/g \lesssim \lambda$. In order for Ipser's approach to be valid, this stringent condition must be satisfied. We do not feel that he has established this to be the case.

Shapiro and Marchant (1978) have recently reported the results of a two-dimensional Monte Carlo simulation of the star distribution around a black hole, in (energy, angular momentum)-space. The angular momentum average of the two-dimensional distribution they obtain is in good agreement with the Bahcall-Wolf (1977) isotropic solution at low binding energy, but exhibits significant depletion at large binding energy. The numerical loss rates found by Shapiro and Marchant (1978) agree well with those of Bahcall and Wolf (1977).

c) The Present Study

The present study was motivated by two basic considerations. First, the assumption that f is nearly isotropic in velocity space has not been established by the previous one-dimensional analyses. (Our computation had already been completed when we learned of the two-dimensional Monte Carlo simulations of Shapiro and Marchant 1978.) Even in the absence of loss cone effects, the $E^{1/4}$ solution does not satisfy the actual steady-state condition (9), but only the weaker condition (14). With inclusion of the loss cone, the dependence of f on J is significant at large binding energy as indicated by equation (63) of Lightman and Shapiro (1977) which shows that, for energies much larger than $E_{\rm crit}$,

$$\frac{f(E, J_{\max})}{f(E, J_{\min})} \gg 1 . \tag{16}$$

Second, we feel that the numerical solution of a fully two-dimensional Fokker-Planck equation including a detailed boundary layer treatment of the loss cone provides a useful check to the loss rates estimated under a number of simplifying assumptions by Lightman and Shapiro (1977) and adopted by Bahcall and Wolf (1977). The λ expansion reduces the kinetic Fokker-Planck equation to an equation in two dimensions (E and J), which is just the degree of complexity that modern Fokker-Planck numerical codes, such as those described by Killeen, Mirin, and Rensink (1976), are capable of handling. Since the finite difference scheme which we use to solve the two-dimensional Fokker-Planck equation is significantly different from the discrete stochastic simulation method recently employed by Shapiro and Marchant (1978), the two computational approaches are complementary. We compare the relative merits of these methods in § VII.

d) An Overview of the Calculation

We present in this paper a numerical solution of the nonlocal Fokker-Planck equation (9). The computation can be separated into the following steps.

1. We first introduce the transformation $(E, J) \rightarrow (E, R)$, where $R \equiv J^2/J_{\max}^2(E)$. The scaled angular momentum R has the convenient property that its allowed range is [0, 1] independent of E. We then write the nonlocal Fokker-Planck equation (9) as a partial differential equation in E and R, which involves the orbit-averaged values of the diffusion coefficients for the phase-space variables E and R. These orbit averages are the time averages of the local values of diffusion coefficients, which depend on r, E, R, and $f(E, R)$, along the orbits specified by E and R. The dependence of the diffusion coefficients on f makes the Fokker-Planck equation nonlinear.
2. We next consider the boundary of the region in (E, R)-space over which the Fokker-Planck equation applies. These boundaries are $R = 1$, the loss cone boundary given by equation (12), and $E = 0$. It is straightforward to derive an appropriate condition for the circular orbit ($R = 1$) boundary. Near the loss cone boundary, however, the λ expansion breaks down, necessitating a boundary layer treatment. The zero energy boundary also involves complications since in this region of (E, R)-space the distribution of stars bound to the black hole merges with that of the unbound stars of the cluster.
3. We next evaluate the term associated with physical collisions, $(\partial f/\partial t)_p$, and obtain the time-averaged physical collision rate along each orbit.
4. Finally, having both the full partial differential equation in (E, R)-space describing the time evolution of f and a complete set of boundary conditions, we choose an initial distribution and advance f in time until a steady state is achieved. The nonlinearity of the Fokker-Planck equation is handled by evaluating the diffusion coefficients using the distribution from the previous time step. We devote a separate section to the discussion of each of these four tasks.

II. THE FOKKER-PLANCK EQUATION

a) Basic Variables

As basic variables in which to write the Fokker-Planck equation we have chosen E and $R \equiv J^2/J_{\max}^2 = 2EJ^2/(GM)^2 = 1 - e^2$, where e is the eccentricity of a bound Keplerian orbit. For any positive E, the allowed

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range of R is [0, 1]. The endpoints of this interval correspond to radial and circular orbits, respectively. The number density $N(E, R)$ in (E, R)-space is related to f by

$$N(E, R)dEdR = \int_{r_-}^{r_+} 4\pi r^2 dr(fd^3v)\,,$$

$$N(E, R) = 2^{1/2}\pi^3(GM)^3E^{-5/2}f_0(E, R)\,, \tag{17}$$

where r_- and r_+ represent the radial limits of the Keplerian orbit specified by (E, R) and are given by

$$r_\pm = \frac{GM}{2E}[1 \pm (1 - R)^{1/2}]\,. \tag{18}$$

Hereafter we suppress the subscript of f_0 for notational simplicity. The orbit-averaged Fokker-Planck equation takes the form

$$\frac{\partial N}{\partial t} = -\frac{\partial}{\partial E}(\langle\Delta E\rangle_t N) + \frac{1}{2}\frac{\partial^2}{\partial E^2}(\langle(\Delta E)^2\rangle_t N) - \frac{\partial}{\partial R}(\langle\Delta R\rangle_t N) + \frac{1}{2}\frac{\partial^2}{\partial R^2}(\langle(\Delta R)^2\rangle_t N)$$

$$+ \frac{\partial^2}{\partial E\partial R}(\langle\Delta E\Delta R\rangle_t N) + \left(\frac{\partial N}{\partial t}\right)_p\,, \tag{19}$$

where we follow Spitzer and Shapiro (1972) in denoting the orbit-average of a local diffusion coefficient by the subscript t. Equation (19) may be obtained directly from equation (9), as demonstrated in the Appendix.

b) Diffusion Coefficients

We compute the local diffusion coefficients using the simplified picture in which the "test" distribution is scattered by an *isotropic* "background" distribution. While the test and background distributions are identical, anisotropies in the background tend to have a smaller effect on the Fokker-Planck operator than those in the test distribution. Thus the assumption of an isotropic background is quite commonly made (cf. Larson 1970; Spitzer and Hart 1971). Under this approximation the distribution we use to calculate the collision integrals is

$$\bar{f}(E) \equiv \int_0^1 dRf(E, R)\,.$$

Shapiro and Marchant (1978) also employ this average.

We note that $\bar{f}(E)$ is not the same distribution as that obtained by averaging the distribution over R at each fixed point r. This latter average depends on both E and r and is not in general a solution of the Vlasov equation. To see the relation between these two mean distributions, consider a radius $r = GM/E$. At this point a background star with energy E must have zero angular momentum, and an average over all angular momenta is inappropriate. For any r and E, R must lie between zero and $R_{\max}(E, r) = 4x(1 - x)$, where $x = rE/GM$. The R average should be taken over this limited range, with the weighting $dR/(R_{\max} - R)^{1/2}$. Since our results indicate that f is monotonic increasing in R for fixed E, it roughly follows that by employing $\bar{f}(E)$ we somewhat overestimate the collision rates for small $R_{\max}$ and underestimate them for large $R_{\max}$.

Integral expressions for the five local diffusion coefficients, the orbit averages of which appear in equation (19), may be obtained by a number of different methods: (1) direct transformation of the Fokker-Planck equation from velocity-space variables as given by Rosenbluth, MacDonald, and Judd (1957) to the coordinates E and R; (2) transformation of the velocity-space diffusion coefficients following the geometrical formalism of Rosenbluth, MacDonald, and Judd (1957; see our Appendix); or (3) the transformation of the (E, J) diffusion coefficients given by Spitzer and Shapiro (1972), generalized to allow for a non-Maxwellian background (see Young 1977*b*). We carried out the derivation of the (E, R) diffusion coefficients by all three methods as an algebraic check. It is convenient to express the results in terms of $x = E/\phi = (E/GM)r$, an energy scaled radial variable, and the following energy integrals:

$$I_0 \equiv 4\pi\Gamma\int_{-\infty}^{E} dE'\bar{f}(E')\,,$$

$$I_{1/2} \equiv 4\pi\Gamma(1 - x)^{-1/2}\int_E^{E/x} dE'\bar{f}(E')\left(1 - x\frac{E'}{E}\right)^{1/2}\,,$$

$$I_{3/2} \equiv 4\pi\Gamma(1 - x)^{-3/2}\int_E^{E/x} dE'\bar{f}(E')\left(1 - x\frac{E'}{E}\right)^{3/2}\,. \tag{20}$$

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Here $\Gamma = 4\pi(Gm)^2 \ln \Lambda$, where m is the stellar mass. Following Spitzer and Hart (1971) we take $\ln \Lambda = \ln (0.4N)$, where N is the number of stars within a radius r_M of the black hole. The local diffusion coefficients are

$$\langle \Delta E \rangle = -I_0 + I_{1/2} ,$$

$$\langle (\Delta E)^2 \rangle = \tfrac{4}{3} E \frac{1-x}{x} I_0 + \tfrac{4}{3} E \frac{1-x}{x} I_{3/2} ,$$

$$\langle \Delta R \rangle = \frac{1}{3E} [8x^2 - 7R] I_0 + \frac{1}{2E} \left[2R - 5R \frac{x}{1-x} + 8x^2 \right] I_{1/2} + \frac{1}{6E} \left[-8R + 3R \frac{x}{1-x} - 8x^2 \right] I_{3/2} ,$$

$$\langle (\Delta R)^2 \rangle = \frac{4}{3} \frac{R}{E} \left[4x^2 - 3R + \frac{R}{x} \right] I_0 + 2 \frac{R}{E} \left[-R \frac{x}{1-x} + 4x^2 \right] I_{1/2} + \frac{2}{3} \frac{R}{E} \left[2R \frac{1-x}{x} - 4R + 3R \frac{x}{1-x} - 4x^2 \right] I_{3/2} ,$$

$$\langle \Delta E \Delta R \rangle = \tfrac{4}{3} R \frac{1-2x}{x} I_0 + \tfrac{4}{3} R \frac{1-2x}{x} I_{3/2} . \tag{21}$$

In order to transform the nonlocal Fokker-Planck equation (9) to (E, R)-space, the five local diffusion coefficients must be averaged over the orbits specified by E and R. For a given R, the energy scaled radial range of the orbit of a star is

$$x_- \le x \le x_+ , \qquad x_\pm = \tfrac{1}{2}[1 \pm (1-R)^{1/2}] . \tag{22}$$

The orbital average of a typical diffusion coefficient $\langle D \rangle$ is given by

$$\langle D \rangle_t = P^{-1} \oint \frac{dr}{v_r} \langle D \rangle = \frac{2}{\pi} \int_{x_-}^{x_+} dx \, x \, Q^{-1/2} \langle D \rangle , \tag{23}$$

where $Q \equiv (x_+ - x)(x - x_-)$. By application of this equation, the following expressions for the orbit-averaged diffusion coefficients are obtained:

$$\langle \Delta E \rangle_t = -F_0 + F_1 ,$$

$$\langle (\Delta E)^2 \rangle_t = \tfrac{4}{3} E F_0 + \tfrac{4}{3} E F_4 ,$$

$$\langle \Delta R \rangle_t = \frac{5}{3E} (1 - 2R) F_0 + \frac{R}{E} F_1 - \frac{5}{2} \frac{R}{E} F_2 + \frac{4}{E} F_3 - \frac{4}{3} \frac{R}{E} F_5 + \frac{R}{2E} F_6 - \frac{4}{3E} F_7 ,$$

$$\langle (\Delta R)^2 \rangle_t = \frac{10}{3} \frac{R}{E} (1 - R) F_0 - 2 \frac{R^2}{E} F_2 + 8 \frac{R}{E} F_3 + \frac{4}{3} \frac{R^2}{E} F_4 - \frac{8}{3} \frac{R^2}{E} F_5 + 2 \frac{R^2}{E} F_6 - \frac{8}{3} \frac{R}{E} F_7 ,$$

$$\langle \Delta E \Delta R \rangle_t = \frac{4R}{3} F_4 - \frac{4R}{3} F_5 , \tag{24}$$

where

$$F_0(E) \equiv 4\pi\Gamma \int_{-\infty}^{E} dE' \bar{f}(E') ,$$

$$F_i(E) \equiv 4\pi\Gamma \int_{E}^{E/x_-} dE' \bar{f}(E') C_i\left(\frac{E'}{E}, R \right) ,$$

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and the functions C_i ($s = E'/E$, R) are given by

$$C_1 \equiv \frac{2}{\pi}\int dx Q^{-1/2}\frac{x(1-sx)^{1/2}}{(1-x)^{1/2}},$$

$$C_2 \equiv \frac{2}{\pi}\int dx Q^{-1/2}\frac{x^2(1-sx)^{1/2}}{(1-x)^{3/2}},$$

$$C_3 \equiv \frac{2}{\pi}\int dx Q^{-1/2}\frac{x^3(1-sx)^{1/2}}{(1-x)^{1/2}},$$

$$C_4 \equiv \frac{2}{\pi}\int dx Q^{-1/2}\frac{(1-sx)^{3/2}}{(1-x)^{1/2}},$$

$$C_5 \equiv \frac{2}{\pi}\int dx Q^{-1/2}\frac{x(1-sx)^{3/2}}{(1-x)^{3/2}},$$

$$C_6 \equiv \frac{2}{\pi}\int dx Q^{-1/2}\frac{x^2(1-sx)^{3/2}}{(1-x)^{5/2}},$$

$$C_7 \equiv \frac{2}{\pi}\int dx Q^{-1/2}\frac{x^3(1-sx)^{3/2}}{(1-x)^{3/2}}, \tag{25}$$

where the limits of integration in each case are x_- and max $[x_-, \min(x_+, s^{-1})]$. These functions represent orbit averages of various powers of the ratio of the velocity of a background star of energy E' to that of a test star of energy E, multiplied by a power of the potential. Numerical evaluation of the C_i on an $(E'/E, R)$ grid is conveniently performed by Chebyshev equal-weight integration. Since the C_i do not involve the distribution function $f(E, R)$, their evaluation need be performed only once for any given grid in $(E'/E, R)$.

c) *Conservation Form of the Fokker-Planck Equation*

To facilitate numerical solution it is useful to recast the Fokker-Planck equation into a particle flux conservation form:

$$\frac{\partial f}{\partial t} = -\mathcal{J}^{-1}\frac{\partial}{\partial E}[\mathcal{J}\phi_E] - \frac{\partial}{\partial R}\phi_R,$$

where

$$\mathcal{J} \equiv 2^{1/2}\pi^3(GM)^3E^{-5/2},$$

$$-\phi_E = D_{EE}\frac{\partial f}{\partial E} + D_{ER}\frac{\partial f}{\partial R} + D_E f,$$

$$-\phi_R = D_{RE}\frac{\partial f}{\partial E} + D_{RR}\frac{\partial f}{\partial R} + D_R f. \tag{26}$$

The quantities $\mathcal{J}\phi_E$ and $\mathcal{J}\phi_R$ are the E-directed flux per unit R and the R-directed flux per unit E, respectively. The six "flux coefficients" appearing above can be expressed in terms of the five (E, R)-space diffusion coefficients:

$$D_{EE} = \tfrac{1}{2}\langle(\Delta E)^2\rangle_t,$$

$$D_{ER} = D_{RE} = \tfrac{1}{2}\langle\Delta E\Delta R\rangle_t,$$

$$D_E = -\langle\Delta E\rangle_t - \frac{5}{4E}\langle(\Delta E)^2\rangle_t + \frac{1}{2}\frac{\partial}{\partial E}\langle(\Delta E)^2\rangle_t + \frac{1}{2}\frac{\partial}{\partial R}\langle\Delta E\Delta R\rangle_t,$$

$$D_{RR} = \tfrac{1}{2}\langle(\Delta R)^2\rangle_t,$$

$$D_R = -\langle\Delta R\rangle_t - \frac{5}{4E}\langle\Delta E\Delta R\rangle_t + \frac{1}{2}\frac{\partial}{\partial E}\langle\Delta E\Delta R\rangle_t + \frac{1}{2}\frac{\partial}{\partial R}\langle(\Delta R)^2\rangle_t. \tag{27}$$

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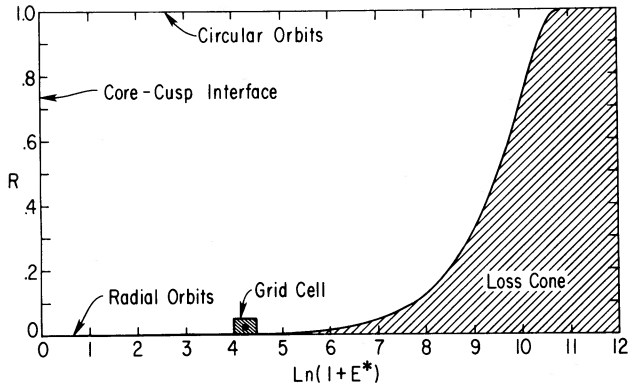


FIG. 1.—Domain of the orbit-averaged Fokker-Planck equation for tidal breakup energy $E_t^* \equiv E_t/\langle v^2\rangle = 5 \times 10^4$. This region of (binding energy, angular momentum)-space is bounded by (1) the core-cusp interface ($E = 0$) where the unbound orbits of the cluster core ($E < 0$) merge with the orbits bound to the black hole ($E > 0$), (2) the circular orbit limit [$R \equiv J^2/J_{\max}^2(E) = 1$], and (3) the loss cone boundary. Orbits in the loss cone have pericenters of less than the tidal breakup radius r_t. For energies $E^* = E/\langle v^2\rangle \lesssim 100$ the loss cone boundary is asymptotic to the radial orbit limit ($R = 0$). A typical cell of the grid used in the finite differencing scheme for integrating the Fokker-Planck equation is indicated.

Using equations (24), (25), and (27) the following expressions are obtained for these six flux coefficients:

$$D_{EE} = \tfrac{2}{3}E(F_0 + F_4)\,,$$

$$D_E = -F_1\,,$$

$$D_{ER} = D_{RE} = \tfrac{2}{3}(F_4 - F_5)\,,$$

$$D_{RR} = \frac{5}{3}\frac{R(1-R)}{E}F_0 - \frac{R^2}{E}F_2 + 4\frac{R}{E}F_3 + \frac{2}{3}\frac{R^2}{E}F_4 - \frac{4}{3}\frac{R^2}{E}F_5 + \frac{R^2}{E}F_6 - \frac{4}{3}\frac{R}{E}F_7\,,$$

$$D_R = -\frac{R}{E}(F_1 - F_2)\,. \tag{28}$$

III. BOUNDARY CONDITIONS

a) The Boundaries

The region of (E, R)-space on which a steady-state solution of the Fokker-Planck equation is to be determined is the rectangular domain indicated in Figure 1. This region is divided by the curve representing the boundary of the loss cone. The part of the rectangle outside the loss cone will be denoted the *main region*. Over most of the rectangle the λ expansion is valid. The breakdown occurs for small R and moderate E. In this region, the loss cone width in R is so small that it does not show on the diagram. The behavior of f in this region is discussed in § III*b*. At the boundary $R = 1$, the flux coefficients D_R, D_{RR}, and D_{RE} vanish, as is to be expected on grounds of symmetry. Thus $\mathcal{J}\phi_R$, the angular flux, vanishes. For this reason no boundary condition, beyond regularity, is needed on this surface.

The $E = 0$ boundary corresponds to that region in phase space where the bound (cusp) stars merge with the unbound (core) stars. The orbits in physical space corresponding to this region of (E, R)-space are so large that they are no longer Keplerian, their outer parts being controlled by the potential of the cluster. We argue in § III*c* that the distribution in this region is close to a Maxwellian so that $f(E, R)$ is equal to the Maxwellian distribution of the unbound stars. We therefore set $f(E_{\min}, R) = (2\pi\langle v^2\rangle)^{-3/2}\exp(E_{\min}/\langle v^2\rangle)$, at some appropriately small $E_{\min}$.

b) The Behavior of f near the Loss Cone

The boundary of the loss cone is given by

$$R_c(E) = 2\frac{E}{E_t}\left(1 - \frac{1}{2}\frac{E}{E_t}\right) \tag{29}$$

which follows from equation (12). R_c represents the fraction of orbits of a given energy which lie in the loss cone. For $E \ll E_t$, this fraction is clearly quite small, although it is unity at $E = E_t$.

Inside the loss cone, stars are lost by tidal breakup in an orbital time scale $P(E)$. For large E, P is quite short

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compared to the collisional diffusion time and stars hardly penetrate beyond the loss cone boundary toward small R. In this case f vanishes throughout the loss cone except for a very small region near the boundary. For small E, P is large and R_c is small so that it is possible for a star to diffuse across the loss cone by small-angle collisions during a single orbital period. Consider a Keplerian orbit in the middle of the loss cone. It intersects the sphere $r = r_t$ so that at this point there can be no stars moving in an outward direction. If we were to follow the orbit outward, we would see stars scattered into it from neighboring orbits. Since stars would also be scattered out of the orbit, an approximate statistical equilibrium would be reached when $r \gg r_t$. Following the orbit past its apocenter and back toward its pericenter, we would again reach the $r = r_t$ sphere and all the inward moving stars in the orbit at this point would be destroyed. More precisely, as we follow the orbit in space, $f^{\rm out}$ (the distribution of outward moving stars) would start from zero at r_t and reach an approximately constant value at some position along the orbit. At the apocenter of the orbit, $f^{\rm out} = f^{\rm in}$ (the distribution of inward moving stars). Following the orbit inward, $f^{\rm in}$ would remain approximately constant until we enter the $r = r_t$ sphere where $f^{\rm in}$ drops discontinuously to zero. Thus in the loss cone, the distribution function $f = \frac{1}{2}(f^{\rm out} + f^{\rm in})$ depends strongly on r as well as E and R. This is also true in the part of the main region which borders the loss cone. For large energies f also varies along an orbit but never reaches a constant value; rather its maximum value, achieved just before the orbit enters the tidal breakup sphere, is exponentially small.

To solve for f in the loss cone and surrounding region, we must return to the space dependent Boltzmann equation (1), dropping the λ expansion of equation (2). However, recalling that the total collision rate is small compared with P^{-1}, we can neglect any change in E during collisions so that E may be treated as a parameter rather than a variable. Equation (1) may be transformed to r, E, R variables:

$$\frac{\partial f}{\partial r} = v_r^{-1} \frac{\partial}{\partial R}\left[\frac{\langle(\Delta R)^2\rangle}{2R} R \frac{\partial f}{\partial R}\right], \tag{30}$$

where $\langle(\Delta R)^2\rangle$ is the local diffusion coefficient given by equation (21) and the other R diffusion coefficients have been neglected since they are comparatively much smaller for $R \ll 1$. For small R, $\langle(\Delta R)^2\rangle/R$ is independent of R but dependent on r (or the scaled variable x). We introduce a new radial variable τ given by

$$\tau = \left[\int_{r_-}^{r} \frac{dr}{v_r} \frac{\langle(\Delta R)^2\rangle}{2R}\right] \Big/ \left[\oint \frac{dr}{v_r} \frac{\langle(\Delta R)^2\rangle}{2R}\right] = \left[P\left(\frac{D_{RR}}{R}\right)_0\right]^{-1} \int_{r_-}^{r} \frac{dr}{v_r} \frac{\langle(\Delta R)^2\rangle}{2R}, \tag{31}$$

the latter equality following from equations (23) and (27). The symbol $(D_{RR}/R)_0$ represents

$$\lim_{R \to 0} (D_{RR}/R).$$

The integral is performed along the orbit; hence τ ranges between 0 and 1 as r varies from r_- to r_+ and back to r_-. Equation (30) becomes

$$\frac{\partial f}{\partial \tau} = P\left(\frac{D_{RR}}{R}\right)_0 \frac{\partial}{\partial R}\left[R \frac{\partial f}{\partial R}\right]. \tag{32}$$

Noting that $(D_{RR}/R)_0$ is independent of τ and defining a new variable y by $R = [P(D_{RR}/R)_0]y$, we obtain the dimensionless equation,

$$\frac{\partial f}{\partial \tau} = \frac{\partial}{\partial y}\left(y \frac{\partial f}{\partial y}\right). \tag{33}$$

Here f is understood to represent $f^{\rm out}$ for $\tau < \frac{1}{2}$ and $f^{\rm in}$ for $\tau > \frac{1}{2}$. Since, for $y < y_c = R_c/[P(D_{RR}/R)_0]$, the orbit is in the loss cone, $f(\tau, y)$ must satisfy

$$f(0, y) = f(1, y) = 0, \qquad y < y_c \tag{34a}$$

$$f(0, y) = f(1, y), \qquad y > y_c. \tag{34b}$$

The second condition follows because outside the loss cone f is a periodic function of τ. Equation (33) must be solved subject to the boundary conditions (34).

It can be seen that τ is a timelike variable along the small R orbits, scaled so that the period of the orbit in τ is unity. From equation (33) we expect a star to diffuse in y a distance $y^{1/2}$ in a single orbital period. (If $y_c \ll 1$, the star will diffuse across the entire loss cone before it is destroyed by the black hole.) Stars with $y < y_c$ will continually be destroyed each period, and in the next period they will be replenished by stars diffusing downward from $y > y_c$. This introduces an oscillation in the distribution function even for $y > y_c$. but this oscillation should die out quickly as y becomes larger than both unity and y_c. Thus, equation (33) can be integrated in τ numerically by setting f constant at $y = y_{\rm max}$, and the resulting solution should be insensitive to the choice of $y_{\rm max}$, for $y_{\rm max}$

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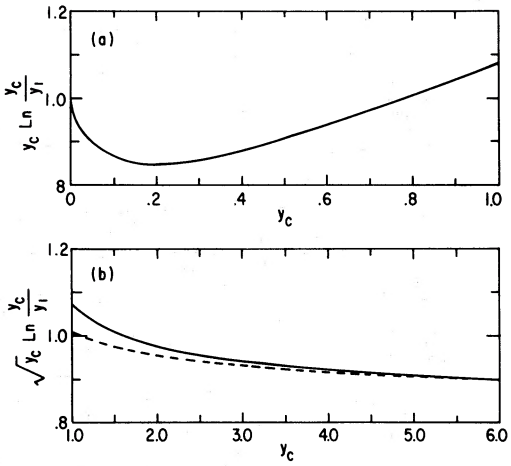


FIG. 2.—The numerical function $\ln (y_c/y_1)$ of y_c which results from the loss cone boundary-layer analysis. Near the loss cone f has the asymptotic form $f \sim c(\ln R/R_c + \ln y_c/y_1)$. In (*a*), $y_c \ln y_c/y_1$ is plotted for $y_c \leq 1$. In (*b*), $y_c^{1/2} \ln y_c/y_1$ is plotted as a solid line and the asymptotic formula $0.824 + 0.186 y_c^{-1/2}$ is plotted as a dashed line, both for $y_c \geq 1$.

large compared to both unity and y_c. This is indeed the case. As expected, we find that such an integration of (33) for several periods of τ, subject to the condition (34a), yields a solution $f(\tau, y)$ which is periodic in τ for all y.

Asymptotically f should be of the form

$$f = c \ln (y/y_1) \tag{35}$$

in the limit $y \gg y_c$, and this form is sufficient for a matching onto the solution in the main region which gives us the correct boundary condition on f_0. This condition can be conveniently expressed in terms of the flux into the loss cone. Making use of equations (26) and (35), we have

$$\begin{aligned} -\phi_R &\approx D_{RR} \frac{\partial f}{\partial R} = \frac{D_{RR}}{R} y \frac{\partial f}{\partial y} \\ &\approx \left(\frac{D_{RR}}{R}\right)_0 \frac{f(y_0)}{\ln (y_0/y_1)} = \left(\frac{D_{RR}}{R}\right)_0 \frac{f(R_0)}{\ln (R_0/R_c) + \ln (y_c/y_1)}, \end{aligned} \tag{36}$$

which gives the flux through the loss cone boundary in terms of $f(R_0)$ for any $R_0 \ll 1$, once $\ln (y_c/y_1)$ is known. Therefore, we express $\ln (y_c/y_1)$ as a function of y_c which we obtain from our numerical integrations of equation (33).

Since we find that $\ln (y_c/y_1)$ is nearly equal to y_c^{-1} for $y_c < 1$, we plot $y_c \ln (y_c/y_1)$ as a function of y_c in this range, in Figure 2*a*. For larger y_c, the solution of equation (33) reduces to a problem treated by Baldwin, Cordey, and Watson (1971) in a study of the plasma distribution in the loss cone of a mirror machine. Their results reduce, in our units, to the *asymptotic* expression

$$\ln \frac{y_c}{y_1} \sim \frac{\zeta(\frac{1}{2})}{(\pi y_c)^{1/2}} = 0.824 y_c^{-1/2} \tag{37}$$

in the limit $y_c \gg 1$. We therefore plot $(y_c)^{1/2} \ln y_c/y_1$ for $y_c > 1$, in Figure 2*b*. We also show the asymptotic formula $0.824 + 0.186 y_c^{-1/2}$ on the same graph.

As discussed by Bahcall and Wolf (1976), the black hole is not expected to remain precisely at the center of the cluster. Instead it will tend to oscillate about the center with rms amplitude

$$\langle r^2 \rangle^{1/2} \approx r_c (m/M)^{1/2}, \tag{38}$$

where r_c is the core radius of the cluster. Young (1977*a*) has considered the effect of this oscillation on the loss cone theory. He concludes that for $E_{\rm crit} < \langle v^2 \rangle$ (where $E_{\rm crit}$ is defined by eq. [54]) the oscillation of the black hole will not influence the loss rate. He argues, however, that when both $E_{\rm crit} > \langle v^2 \rangle$ and $\langle r^2 \rangle^{1/2} \gg b_t \equiv (2GMr_t/3\langle v^2 \rangle)^{1/2}$, the loss-cone width in J will tend to be smeared out by the ratio $\langle r^2 \rangle^{1/2}/b_t$. He finds that even if these two conditions are simultaneously satisfied (which is not the case for a globular cluster), the loss rates are not drastically altered. In addition it might be noted that when the Keplerian period P of a star about the black hole is much shorter than the harmonic oscillation period $P_H \approx 2r_c/\langle v^2 \rangle^{1/2}$ of the black hole in the cluster core, the adiabatic approximation should apply and the stellar orbit will remain essentially Keplerian in the weakly noninertial frame of

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the black hole. It is appropriate to evaluate the ratio P/P_H at the energy, $E_{\rm crit}$, at which the flux of stars into the loss cone peaks; we find

$$\frac{P}{P_H} \approx \frac{r_h}{r_c}\left(\frac{\langle v^2\rangle}{E_{\rm crit}}\right)^{3/2}, \tag{39}$$

where $r_h \equiv GM/\langle v^2\rangle$. For $E_{\rm crit} \gg \langle v^2\rangle$ and for $r_h \lesssim r_c$, $P \ll P_H$. Thus in any event the approximation that the black hole is fixed in position should not lead to serious errors in the loss rate.

c) *The Small Energy Region*

The last boundary we must discuss is that near $E = 0$. For stars with $E \leq GM/r_M$ the orbits are sufficiently large that they no longer may be considered to be solely determined by the gravitational field of the black hole. For such orbits our procedure should strictly be generalized, the orbit averages of the diffusion coefficients being carried out over the correct non-Keplerian orbits. However, these stars have velocities which are small compared with the mean velocities in the cluster core, so small-angle collisions will change their orbits appreciably during even a single period. Thus, in this regime the λ expansion is again unreasonable. Nevertheless, it seems plausible that the distribution is nearly Maxwellian in this region; this is verified by our numerical calculations. The deviations from a Maxwellian distribution may be considered to be produced by any net flux present in phase space. Such a flux is of course present in our problem since the stars which are consumed must, in a steady state, be replaced by stars which are captured from the cluster core. If this flux can be shown to be small, however, we can be reasonably certain that the distribution is Maxwellian. The actual number of stars in the region of radius r_M inside which the gravitational force of the black hole dominates that of the cluster is of order the ratio M/m, which is by assumption $\sim 10^3$. The flux produced by a significant deviation from a Maxwellian would correspond to all of the stars in this sphere being lost in a relaxation time which, for a typical globular cluster, core is $\sim 10^8$ yr. In contrast, our numerical results indicate that for a $10^3\ M_\odot$ black hole only about 30 stars are lost per 10^8 yr. Thus the relative smallness of the actual flux indicates that the distribution must be close to a Maxwellian for the unbound and weakly bound stars.

Having made plausible the near Maxwellian nature of the distribution function, let us turn to the question of what boundary condition to apply on f near the $E = 0$ surface. We have shown that the Fokker-Planck equation does not strictly apply in the region of small E. However, arguments similar to the above indicate that near $E = 0$ the solution of our Fokker-Planck equation must again be a Maxwellian with a temperature characteristic of the temperature of the core of the cluster. By matching the solution at any sufficiently small energy to an appropriately normalized Maxwellian we should get a distribution function which agrees with that for the true situation. We take the energy for our matching as zero and set

$$f(0, R) = n(2\pi\langle v^2\rangle)^{-3/2} \tag{40}$$

as our boundary condition on this surface.

IV. STELLAR COLLISIONS

As pointed out by previous authors, within a radius $r_{\rm coll}$ (defined by $GM/r_{\rm coll} \approx Gm/a$) the impact parameter for a 90° deflection becomes comparable with the radius a of a star. Stars cannot diffuse to energies much larger than $E_{\rm coll} \equiv GM/r_{\rm coll}$ since actual physical collisions dominate energy scattering above this energy. Since the relative velocity of two colliding stars of $E > E_{\rm coll}$ is typically larger than the escape velocity from the stellar surfaces, collisions lead to stellar disruption and thus loss from the distribution. On this assumption, the numerical studies of Bahcall and Wolf (1976, 1977) have taken $f(E_{\rm coll}) = 0$ as a boundary condition. Shapiro and Marchant (1978) have similarly taken f to vanish at the larger of $r_{\rm coll}$ and r_t.

In order to study the effect of star loss due to collisions in more detail, we consider the physical collision cross section for stars of radius a,

$$\sigma_p(v_{\rm rel}) = \pi(2a)^2(1 + v_{\rm esc}^2/v_{\rm rel}^2), \tag{41}$$

where $v_{\rm esc} = (2Gm/a)^{1/2}$ is the escape velocity from the stellar surface and $v_{\rm rel}$ is the relative collision velocity at infinity. The collision rate $\nu_{\rm coll}$ is obtained by integrating $f\sigma_p v_{\rm rel}$ over all velocities of background particles:

$$\nu_{\rm coll} = \int d^3v' f(v')\sigma_p(|\boldsymbol{v} - \boldsymbol{v}'|)|\boldsymbol{v} - \boldsymbol{v}'| = 4\pi a^2 g(v) + 4\pi Gmah(v), \tag{42}$$

where $g(v)$ and $h(v)$ are the "Rosenbluth potentials" given by equation (A3).

The collision rate given by equation (42) is clearly a function of position along a stellar orbit. A time-averaged collision rate may be obtained by averaging equation (42) over stellar orbits following the procedure of § II*b*. We find

$$\nu_{{\rm coll},t} = 4\pi a^2 g_t + 4\pi Gmah_t, \tag{43}$$

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where

$$g_t = \frac{8\pi}{3} E \int_{-\infty}^{E} dE' \bar{f}(E') \left[7 - 3\frac{E'}{E}\right] + \frac{8\pi}{3} E \int_{E}^{E/x_-} dE' \bar{f}(E') \left[3\left(\frac{E'}{E} - 1\right) C_1 + 4C_4\right],$$

$$h_t = 8\pi \int_{-\infty}^{E} dE' f(E') + 8\pi \int_{E}^{E/x_-} dE' f(E') C_1;$$

and the functions C_1 and C_4 are defined by equation (25).

On the assumption that physical collisions result in star destruction, the physical collision loss term in equation (9) is given by

$$\left(\frac{\partial f}{\partial t}\right)_p = -\nu_{\mathrm{coll},t} f. \tag{44}$$

In fact, the low-velocity collisions ($v_{\mathrm{rel}} < v_{\mathrm{esc}}$) of weakly bound stars ($E \lesssim \langle v^2 \rangle$) probably result in stellar coalescence rather than disruption (Spitzer and Saslaw 1966; Colgate 1967; Sanders 1970). However, the binary encounter rate for weakly bound stars strongly dominates the physical collision rate. Thus the details of the treatment of physical collisions have little effect on f at low binding energies. At large binding energies, where physical collisions are dominant, the average collision velocity v_{rel} is sufficiently large to ensure star destruction.

V. NUMERICAL METHOD

a) *Choice of Units*

The Fokker-Planck equation may be written in a dimensionless form by introducing a dimensionless energy E^*, distribution function f^*, and time t^* defined by

$$E = \langle v^2 \rangle E^*, \qquad f = n(2\pi\langle v^2 \rangle)^{-3/2} f^*, \qquad t = n^{-1}(2\pi\langle v^2 \rangle)^{3/2} \Gamma^{-1} t^*, \tag{45}$$

where $\langle v^2 \rangle$ is the *one-dimensional* velocity dispersion in the cluster core and n is the core density. The time unit defined by this scaling is

$$t_0 = 2.2 \times 10^8 \text{ yr} \left(\frac{n}{5 \times 10^4 \text{ pc}^{-3}}\right)^{-1} \left(\frac{\langle v^2 \rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right) \left(\frac{m}{M_\odot}\right)^{-2} = 3.7 t_{rh}, \tag{46}$$

where a value of 6 has been taken for $\ln \Lambda$ and t_{rh} is the reference relaxation time introduced by Spitzer and Hart (1971). In terms of the dimensionless energy, the Keplerian period of a stellar orbit is

$$P^* E^{*3/2} = 4.3 \times 10^{-5} \left(\frac{M}{10^3 M_\odot}\right) \left(\frac{n}{5 \times 10^4 \text{ pc}^{-3}}\right) \left(\frac{\langle v^2 \rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right)^{-3} \left(\frac{m}{M_\odot}\right)^2. \tag{47}$$

Dimensionless forms for the flux coefficients and the Rosenbluth potentials are obtained by replacing Γ by unity, E by E^*, and f by f^* in equations (28) and (43). The dimensionless collision rate is given by

$$\nu^*_{\mathrm{coll},t} = 4.6 \times 10^{-8} \left(\frac{\langle v^2 \rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right)^2 \left(\frac{m}{M_\odot}\right)^{-2} \left(\frac{a}{R_\odot}\right)^2 g_t^* + 8.8 \times 10^{-5} \left(\frac{\langle v^2 \rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right) \left(\frac{m}{M_\odot}\right)^{-1} \left(\frac{a}{R_\odot}\right) h_t^*, \tag{48}$$

where g_t^* and h_t^* are the dimensionless orbit-averaged Rosenbluth potentials.

The dimensionless tidal breakup energy is given by

$$E_t^* = \frac{GM}{2r_t \langle v^2 \rangle} = 4.7 \times 10^4 \left(\frac{M}{10^3 M_\odot}\right)^{2/3} \left(\frac{\langle v^2 \rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right)^{-1} \left(\frac{m}{M_\odot}\right)^{1/3} \left(\frac{a}{R_\odot}\right)^{-1} \tag{49}$$

where the value of r_t given by Frank and Rees (1976) has been used. We note that Bahcall and Wolf (1977) take $E_t^* = 1.7 \times 10^5$ independent of the black hole mass.

For any $E_0 > 0$ the total number of stars bound to the black hole with binding energy greater than E_0 is

$$N = 11 \left(\frac{M}{10^3 M_\odot}\right)^3 \left(\frac{n}{5 \times 10^4 \text{ pc}^{-3}}\right) \left(\frac{\langle v^2 \rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right)^{-3} \int_{E_0/\langle v^2 \rangle}^{E_t^*} dE^* E^{*-5/2} \int_0^1 dR f^*(E^*, R). \tag{50}$$

b) *Basic Numerical Approach*

Because of the large energy range under consideration ($E_t^* \approx 5 \times 10^4$ for $M = 10^3 M_\odot$), we replace E^* by a logarithmic variable $Z = \ln(1 + \beta E^*)$. A value of 5 was chosen for the scaling parameter β in order to provide an adequate representation of the distribution function for $E^* < 1$, on a uniform Z mesh.

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The region of (E, R)-space on which the Fokker-Planck equation is to be solved is represented by a uniform grid of $N_Z \times N_R$ cells, one of which is illustrated in Figure 1. The distribution function f is evaluated at cell centers while the flux coefficients and fluxes are evaluated at cell boundaries. The rate of change of f in a cell is then equated to the sum of all the inward flux through the cell boundary minus any loss term. The dimensionless evolution equation,

$$\frac{\partial f^*}{\partial t^*} = -\mathcal{J}^{*-1}\beta e^{-z}\frac{\partial}{\partial Z}[\mathcal{J}^*\phi_E^*] - \frac{\partial}{\partial R}\phi_R^* - \nu^*_{\mathrm{coll},t} f^*, \tag{51}$$

was written in implicit discrete form, with the time derivative represented as a backwards difference and the flux coefficients evaluated at the previous time step. The set of $N_Z \times N_R$ difference equations then takes the form

$$\frac{f_{ij}^{(n+1)} - f_{ij}^{(n)}}{\Delta t} = \sum_{k=i-1}^{i+1}\sum_{l=j-1}^{j+1} A_{ijkl}^{(n)} f_{kl}^{(n+1)}, \tag{52}$$

where $f_{ij}^{(n)} = f(Z_i, R_j)$ evaluated at the nth time step. The integration of (52) in time (i.e., the determination of $f^{(n+1)}$ given $f^{(n)}$) requires the inversion of a matrix of rank $N_Z \times N_R$ and bandwidth about the main diagonal of order 2 min (N_Z, N_R). This proved entirely tractable for a grid of size $(N_Z = 25) \times (N_R = 20)$. In contrast to the more common alternating-direction implicit method of solution of two-dimensional partial differential diffusion equations, the method employed here permits the use of a large time step, $\Delta t \gg t_0$. In the limit of infinite time step the method resembles the relaxation techniques used in the study of equilibrium stellar structure; the convergence to steady-state solutions is rapid. Since the flux coefficients depend on f itself, the A's in equation (52) must be recomputed at each step. Thus several time steps are necessary to achieve a steady-state solution.

The numerical code used to integrate equation (52) in time is based on one originally developed by Killeen and Marx (1970) for studies of collisional plasmas and subsequently improved by Kulsrud *et al.* (1973). The accuracy of the present code was tested by integrating an angular momentum averaged Fokker-Planck equation equivalent to equation (44) of Bahcall and Wolf (1976). The agreement with the solution presented by Bahcall and Wolf (1977; Table 1) is better than 4% at all energies.

c) Numerical Treatment of the Loss Cone

The loss cone boundary condition derived in § III*b* can be expressed either in terms of the required behavior of f near the loss cone boundary (eq. [35]) or in terms of the flux of stars into the loss cone (eq. [36]). When $R_c(E) \ll \Delta R$ (the width of a grid cell), the latter form is more convenient.

Consider the grid cell indicated in Figure 1; a small fraction of this cell lies within the loss cone. Since the loss cone boundary in this cell is nearly coincident with the $R = 0$ boundary, the flux into the loss cone may be represented, for computational purposes, as an R-flux through $R = 0$. The magnitude of this flux is computed from the value of f at the cell center using equation (36). For $R_c \geq \Delta R/2$ (we have taken $\Delta R = 0.05$) it is also the case that $\ln(y_c/y_1) \ll 1$ for typical globular cluster parameters. Thus it follows from equation (35) that the loss cone boundary condition can be simplified to

$$f(E, R_c) = 0 \tag{53}$$

when $E \geq 0.01 E_t$.

We note that Ipser (1978*a*, *b*) applies this boundary condition at *all* energies. We have numerically investigated this treatment of the loss cone boundary and find that the resultant energy profile of the steady-state distribution is depleted relative to that produced by our detailed boundary layer treatment.

VI. RESULTS

Starting from an arbitrary initial form, the stellar distribution function was advanced in time until a steady state was achieved. This was done for several sets of values of the parameters, M, n, and $\langle v^2 \rangle$. For $\Delta t = 10 t_{rh}$, about three time steps were found to be sufficient for this purpose, even when the adopted initial distribution was quite different from the steady-state solution. With a time step this large, the numerical method produces a solution of the steady-state Fokker-Planck equation by successive relaxation of approximate solutions. Trials using a step size $\Delta t \ll t_{rh}$ indicate that the physical time scale for achieving a steady state is $\sim t_{rh}$.

Figure 3 illustrates the resulting two-dimensional distribution function for the standard globular cluster parameters considered by previous authors: $M = 10^3\ M_\odot$, $n = 5 \times 10^4\ \mathrm{pc}^{-3}$, $\langle v^2 \rangle = 100\ \mathrm{km}^2\ \mathrm{s}^{-2}$. The natural logarithm of f^* is plotted as a surface over the (E^*, R)-plane. As expected from the discussion of § III*c*, f is isotropic and Maxwellian near the zero energy boundary. As the binding energy E increases, f becomes increasingly anisotropic due to the growing influence of the loss cone. The R profile of f near the loss cone for $\ln(1 + E^*) < 5$ does not show since f is only plotted over the adopted (E, R)-grid, viz., for R in the range $0.025 \leq R \leq 0.975$.

Figure 4 illustrates the R averaged distribution $\bar{f}(E)$ for black hole masses of 10^2, 10^3, and $10^4\ M_\odot$ and a value n and a value $\langle v^2 \rangle$ appropriate to centrally concentrated globular clusters. The dimensionless energies E_{crit}^*, E_{coll}^*, and E_t^* for these models are listed in Table 1, together with the numerical values for the rate of tidal destruction of stars and the rate of physical collisions.

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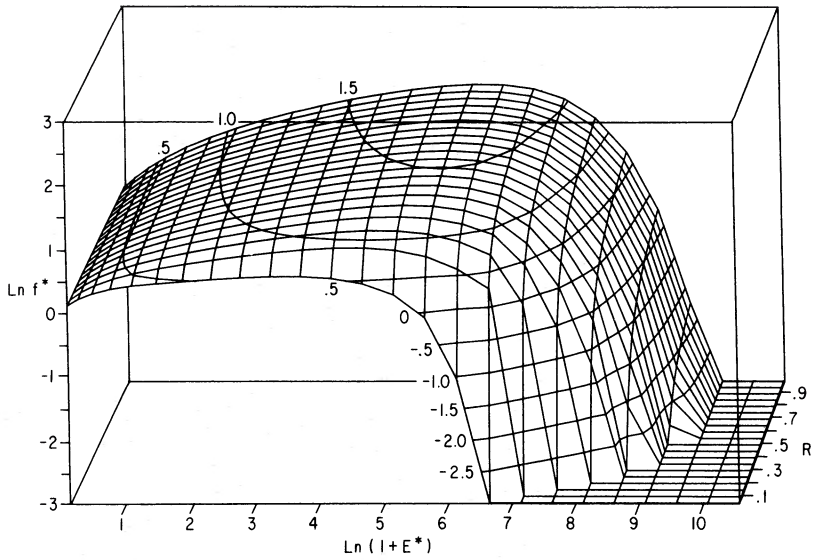


FIG. 3.—The natural logarithm of the normalized two-dimensional distribution function f^* plotted as a surface over the (E^*, R)-plane $[E^* \equiv E/\langle v^2 \rangle, R \equiv J^2/J_{\max}^2(E)]$. Coordinate lines of constant E^* and R are drawn on the surface. Contours of constant $\ln f^*$ are indicated. The distribution is not shown for $\ln f^* < -3$. The model parameters corresponding to this distribution are: black hole mass $M = 10^3\ M_\odot$, cluster core density $n = 5 \times 10^4$ stars pc^{-3}, and core velocity dispersion $\langle v^2 \rangle = 100$ km^2 s^{-2}. Note the flat R gradient for $E^* \sim 1$ where the distribution joins the isotropic core distribution. The increasing depletion of small R orbits with increasing E^* reflects the growing importance of the loss cone.

The perturbation in $\bar{f}$ produced by inclusion of loss cone effects is larger in the present study than that found by Bahcall and Wolf (1977, Fig. 1). They report that the perturbation Δp in $p(E^*) = d \ln f^*/d \ln E^*$ satisfies the bound $|\Delta p| \leq 0.02$ for $\ln (1 + E^*) \leq 4$, while we find that $|\Delta p|$ is as large as 0.2 in this energy range, when $M = 10^2\ M_\odot$. The differences between our results and those of Bahcall and Wolf (1977) are smaller for more massive black holes (i.e., $M \approx 10^3$–$10^4\ M_\odot$), but in all cases are noticeable for large E^*. A comparison between our results and those of Bahcall and Wolf (1977) and Shapiro and Marchant (1978) is shown in Figure 5. The models of Shapiro and Marchant (1978) are parametrized by the dimensionless energies E_t^* and $E_{\rm crit}^*$, where

$$E_{\rm crit}^* = 4.7 \left(\frac{M}{10^3\ M_\odot}\right)^{20/27} \left(\frac{n}{5 \times 10^4\ \text{pc}^{-3}}\right)^{4/9} \left(\frac{\langle v^2 \rangle}{100\ \text{km}^2\ \text{s}^{-2}}\right)^{-16/9} \left(\frac{m}{M_\odot}\right)^{28/27} \left(\frac{a}{R_\odot}\right)^{-4/9} \tag{54}$$

is the energy where loss cone effects become important. For the purpose of comparison the model shown in Figure 5 was computed for $E_t^* = 10^4$, $E_{\rm crit}^* = 10$, and with the physical collision cross section set to zero. The agreement among all three studies is reasonable for $E^* \lesssim 50$, while for larger energies both our distribution and that of Shapiro and Marchant (1978) are significantly depleted below the Bahcall and Wolf (1977) result. This difference

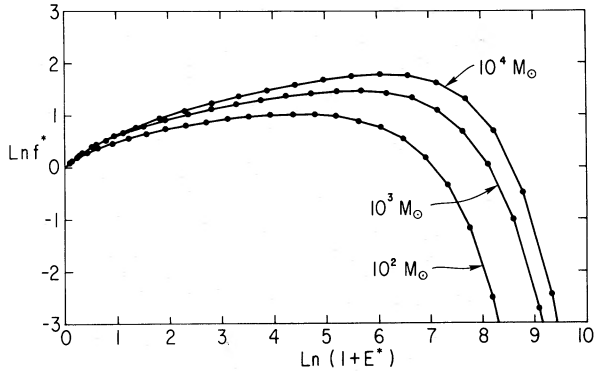


FIG. 4.—Angular momentum averaged distribution functions. The black hole mass associated with each curve is indicated. The cluster core parameters are $n = 5 \times 10^4$ stars pc^{-3} and $\langle v^2 \rangle = 100$ km^2 s^{-2}. The distributions are normalized so that they smoothly match the unbound distribution $f_u^* = \exp(E^*)$ at $E^* = 0$.

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TABLE 1

NUMERICAL RATES

M ($M_\odot$)	n (pc^{-3})	$\langle v^2\rangle$ (km^2 s^{-2})	E^*_{crit}	E^*_{coll}	E_t^*	$\dot{N}_t$ (10^{-8} yr^{-1})†	$\dot{E}/\dot{N}_t\langle v^2\rangle$‡	$\dot{N}_p$ (10^{-8} yr^{-1})§
10^2.....	5×10^4	100	0.85	2×10^3	10^4	0.21	1.5	8.8×10^{-4}
10^3.....	5×10^4	100	4.7	2×10^3	5×10^4	30	9.1	0.89
10^4.....	5×10^4	100	26	2×10^3	2×10^5	5.4×10^3	42	9.2×10^2
10^3.....	10^4	100	2.3	2×10^3	5×10^4	2.6	4.1	3.4×10^{-2}
10^3.....	2.5×10^5	100	9.6	2×10^3	5×10^4	3.8×10^2	18	23
10^3.....	5×10^4	25	55	8×10^3	2×10^5	1.4×10^3	1.0×10^2	1.2×10^2
10^3.....	5×10^4	400	0.40	5×10^2	10^4	1.0	0.61	6.5×10^{-3}

† Rate of tidal destruction.

‡ Average energy of tidally destroyed stars.

§ Physical collision rate ($E \geq 0.05\langle v^2\rangle$).

is not surprising, since the analysis of Bahcall and Wolf (1977) neglects the angular momentum dependence of the distribution function which becomes significant at large energies due to the importance of the loss cone.

Figure 6 shows constant energy R-profiles of the two-dimensional distribution function illustrated in Figure 3. It is clear that at all energies $f(E, R)$ is closely approximated by the analytic form

$$f(E, R) \approx A(E) + B(E) \ln R \tag{55}$$

over the *entire* range $R_c \leq R \leq 1$. The flattening of the profiles for large energies ($E^* \geq 5000$) is a consequence of star loss due to physical collisions.

a) Loss Rates

The scaling of the loss rates listed in Table 1 with the parameters M, n, and $\langle v^2\rangle$ agrees reasonably with the scaling law derived by Frank and Rees (1976, eq. [16a]), although our numerical values are about 6 times larger. For the "standard" globular cluster parameters ($M = 10^3\ M_\odot$, $n = 5 \times 10^4$ pc^{-3}, $\langle v^2\rangle = 100$ km^2 s^{-2}) we find a tidal destruction rate of 30 stars per 10^8 yr, which scales as $M^{2.3} n^{1.6} \langle v^2\rangle^{-2.9}$ (see eq. [66] below). Our loss rate is a factor of 15 larger than estimated by Lightman and Shapiro (1977, Table 2), about 3 times larger than the numerical rate of Bahcall and Wolf (1977, eq. [16]), and twice as large as found by Shapiro and Marchant (1978, eq. [31]). The discrepancy between our loss rate and that of Shapiro and Marchant (1978) is unexpectedly large considering the good agreement of our distribution functions (see Fig. 5). It evidently arises from differences in our mathematical treatment of the loss cone.

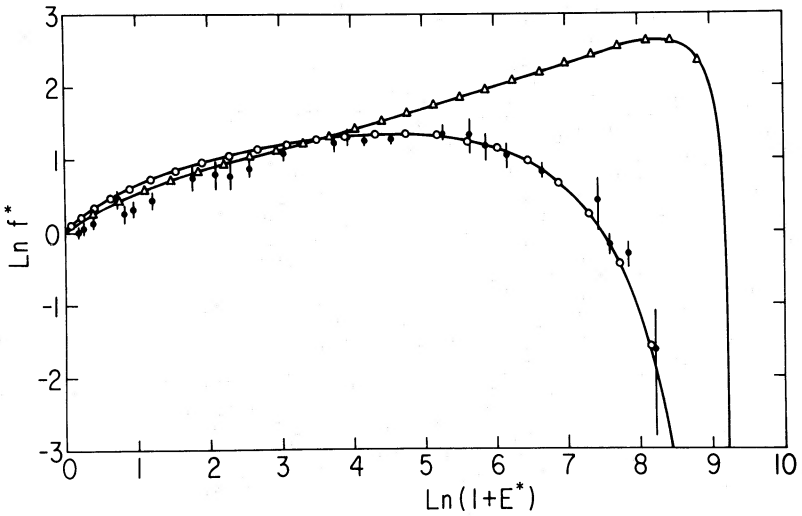


FIG. 5.—Comparison of angular momentum averaged distribution functions: *open circles*, present study; *filled circles*, Shapiro and Marchant 1978, Table 1; *triangles*, Bahcall and Wolf 1977, Fig. 1*b*. For the first two studies $E_{crit}^* = 10$ and $E_t^* = 10^4$. For the third study $E_t^* \approx 2 \times 10^5$; however, the distribution was forced to vanish for $E^* \geq 10^4$. The physical collision cross section was set to zero in the present study for this comparison, since physical collisions are not explicitly included by Shapiro and Marchant (1978) or by Bahcall and Wolf (1977).

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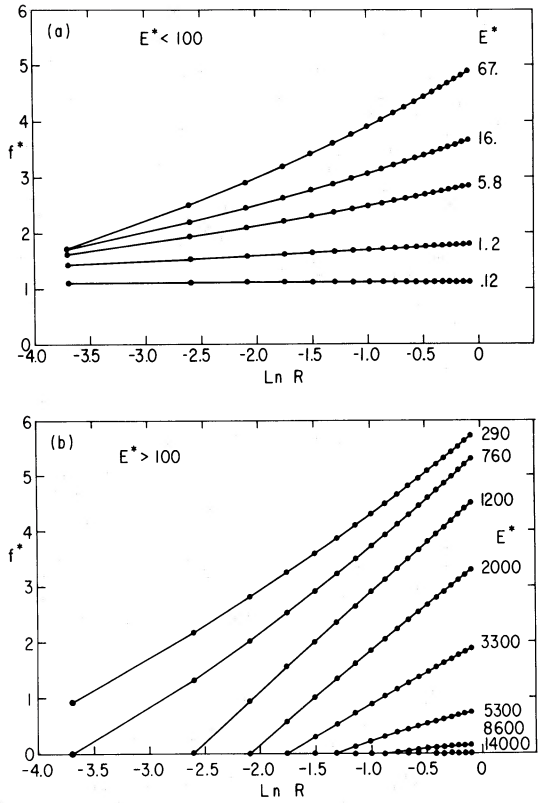


FIG. 6.—Constant energy angular momentum profiles of the distribution function of Fig. 3. The dimensionless energy $E^* \equiv E/\langle v^2 \rangle$ associated with each curve is indicated. Note that all curves are reasonably approximated by $f^* \sim A(E^*) + B(E^*) \ln R$.

We find that the physical collision rate in the cusp obeys the scaling law,

$$\dot{N}_p = 0.9 \times 10^{-8}\ \mathrm{yr}^{-1} \left(\frac{M}{10^3\ M_\odot}\right)^3 \left(\frac{n}{5 \times 10^4\ \mathrm{pc}^{-3}}\right)^2 \left(\frac{\langle v^2 \rangle}{100\ \mathrm{km}^2\ \mathrm{s}^{-2}}\right)^{-7/2} . \tag{56}$$

(The cusp is somewhat arbitrarily defined here as those stars bound to the black hole with energy $E \geq 0.05\langle v^2 \rangle$.) This rate, which includes both low- and high-velocity collisions, is in good agreement with Young's (1977*a*) estimate (see his eq. [A9d]). As noted by Young, low-velocity collisions ($v_{\mathrm{rel}} < v_{\mathrm{esc}}$; see § IV) are more numerous in the cusp than high-velocity collisions ($v_{\mathrm{rel}} > v_{\mathrm{esc}}$), and occur mostly in the outer part of the cusp ($E \leq \langle v^2 \rangle$). For $M = 10^3\ M_\odot$, $n = 5 \times 10^4\ \mathrm{pc}^{-3}$, and $\langle v^2 \rangle = 100\ \mathrm{km}^2\ \mathrm{s}^{-2}$, about 70% of the physical collisions occur below the energy $\langle v^2 \rangle$. Although we assumed that all physical collisions resulted in stellar destruction, this simplification was found to have little effect on the low-energy behavior of the distribution function. This result is expected since the loss cone flux exceeds the physical collision rate by factors of 240, 30, and 6 for black holes of 10^2, 10^3, and $10^4\ M_\odot$, respectively, and the standard values of n and $\langle v^2 \rangle$. For much more massive black holes, however, it would be necessary to distinguish between low- and high-velocity collisions in order to accurately determine the low-energy profile of the distribution function. It should be noted that since we regard the potential as Keplerian even for $r \sim 10GM/\langle v^2 \rangle$, which is comparable with the cluster core radius in some cases listed in Table 1, our physical collision rates may be too large by about a factor of 2 in these cases.

The scaling of the loss rate can be derived in a straightforward way. As shown in § III*b*, the flux into the loss cone can be written in terms of the value of $\partial f/\partial R$ at some point near the loss cone boundary. Our numerical results indicate that f closely follows the analytic form,

$$f(E, R) = A(E)\left[\ln \frac{y_c}{y_1} + \ln \frac{R}{R_c}\right], \tag{57}$$

for all $R \geq R_c$. We recall that $y_c = [P(D_{RR}/R)_0]^{-1} R_c$ and that $\ln (y_c/y_1) \approx y_c^{-1}$ for $y_c \leq 1$. The R-average of f is

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easily computed:

$$\bar{f}(E) \approx \int_{Rc}^{1} dR f(E, R) \approx A(E)\left[\ln \frac{y_c}{y_1} + \ln R_c^{-1} - 1\right] \tag{58}$$

for $R_c \ll 1$. The flux into the loss cone per unit energy interval is given by

$$-(\mathscr{J}\phi_R)_0 \approx \mathscr{J}\left(\frac{D_{RR}}{R}\right)_0 R\frac{\partial f}{\partial R}. \tag{59}$$

Using equations (57) and (58), we find

$$R\frac{\partial f}{\partial R} \approx \frac{\bar{f}(E)}{\ln (y_c/y_1) + (\ln R_c^{-1} - 1)}, \tag{60}$$

which is independent of R. Neglecting scattering by stars with energy larger than that of the test star, we have

$$\left(\frac{D_{RR}}{R}\right)_0 \approx \frac{20\pi\Gamma}{3E}\int_{-\infty}^{E} dE'\bar{f}(E'). \tag{61}$$

The total rate of loss of stars to the loss cone is given by the integral,

$$\begin{aligned}\dot{N} &= \int_0^\infty dE(-\mathscr{J}\phi_R)_0 \\ &\approx 1.1 \times 10^{-6}\,\mathrm{yr}^{-1}\left(\frac{M}{10^3\,M_\odot}\right)^3\left(\frac{n}{5\times 10^4\,\mathrm{pc}^{-3}}\right)^2\left(\frac{\langle v^2\rangle}{100\,\mathrm{km^2\,s^{-2}}}\right)^{-9/2} \\ &\quad\times\left(\frac{m}{M_\odot}\right)^2\int_0^\infty dE^*\,\frac{E^{*-7/2}\bar{f}^*(E^*)}{\ln (y_c/y_1) + (\ln R_c^{-1} - 1)}\int_{-\infty}^{E^*} dE'^*\bar{f}^*(E'^*).\end{aligned} \tag{62}$$

Consider the two terms in the denominator of the integrand. At small energies ($E^* \ll 1$), $\ln (y_c/y_1) \gg \ln R_c^{-1}$; conversely, at large energies ($E^* \gg 1$), $\ln (y_c/y_1) \ll \ln R_c^{-1}$. The two terms are of the same magnitude when $y_c \approx 1/10$. Thus for the purpose of evaluating the integral in equation (62), the approximation $\ln (y_c/y_1) \approx y_c^{-1}$ is adequate for all energies. As shown by Bahcall and Wolf (1976), the dimensionless distribution function $f^*(E^*)$ is reasonably approximated by the analytic form:

$$\begin{aligned} f^*(E^*) &\sim \exp(E^*) \quad (E^* \lesssim 0.5) \\ &\sim 2E^{*1/4} \quad (0.5 \lesssim E^* \lesssim E_{\mathrm{coll}}^*) \\ &\sim 0 \quad (E_{\mathrm{coll}}^* \lesssim E^*). \end{aligned} \tag{63}$$

Using these expressions, the loss rate integral appearing in equation (62) reduces to

$$I_1 = 2\int_0^{E^*_{\mathrm{coll}}} dE^*\,\frac{E^{*-13/4}(1 + 1.6E^{*5/4})}{[\ln (E_t^*/2) - \ln E^* - 1] + AE^{*-7/2}(1 + 1.6E^{*5/4})}, \tag{64}$$

where

$$\begin{aligned} A &= 21\left(\frac{M}{10^3\,M_\odot}\right)^{5/3}\left(\frac{n}{5\times 10^4\,\mathrm{pc}^{-3}}\right)\left(\frac{\langle v^2\rangle}{100\,\mathrm{km^2\,s^{-2}}}\right)^{-4}\left(\frac{m}{M_\odot}\right)^{7/3}\left(\frac{a}{R_\odot}\right)^{-1} \\ &= 0.65E_{\mathrm{crit}}^{*9/4}. \end{aligned}$$

It is also useful to define an energy loss rate integral I_2 which differs from I_1 by an extra factor of E^* in the integrand.

We have evaluated I_1 and I_2 numerically as functions of the parameters A and E_{coll}^* for $E_t^* = 4.7 \times 10^4$. As can be seen from (64), I_1 and I_2 depend on E_{coll}^* rather weakly. While I_1 is essentially independent of E_{coll}^* for $E_{\mathrm{coll}}^* \gg 1$, I_2 depends logarithmically on E_{coll}^*. For $E_{\mathrm{coll}}^* = 10^3$ we find that our numerical results for A in the range $1 \le A \le 10^3$ are well fitted by the following analytic forms:

$$\begin{aligned} I_1 &\approx 0.958A^{-0.404}, \\ I_2 &\approx 4.08A^{-0.062}. \end{aligned} \tag{65}$$

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Thus we obtain the scaling laws:

$$\dot{N} = 30(10^8 \text{ yr})^{-1}\left(\frac{M}{10^3\,M_\odot}\right)^{2.33}\left(\frac{n}{5\times 10^4 \text{ pc}^{-3}}\right)^{1.60}\left(\frac{\langle v^2\rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right)^{-2.88}\left(\frac{m}{M_\odot}\right)^{1.06}\left(\frac{a}{R_\odot}\right)^{0.40},$$

$$\frac{\dot{E}}{\dot{N}\langle v^2\rangle} = 12\left(\frac{M}{10^3\,M_\odot}\right)^{0.57}\left(\frac{n}{5\times 10^4 \text{ pc}^{-3}}\right)^{0.34}\left(\frac{\langle v^2\rangle}{100 \text{ km}^2 \text{ s}^{-2}}\right)^{-1.37}\left(\frac{m}{M_\odot}\right)^{0.80}\left(\frac{a}{R_\odot}\right)^{-0.34}. \quad (66)$$

The first scaling law is in excellent agreement with our detailed numerical results for the standard globular cluster case, while the agreement of the second is not quite as good. This is not surprising, since I_1 is essentially independent of the large energy behavior of f^* where the deviations from the approximate form (63) become significant. Because of the extra factor of E^* in the integrand of I_2, the value of this integral depends more sensitively on the large energy behavior of f^*.

b) The Run of Density and Velocity Dispersion

At a radius r, the stellar density $n(r)$ is given by

$$n(r) = \int d^3v f$$
$$= \pi^{-1/2} n_0 \phi^{*3/2} \int_0^1 dx(1-x)^{1/2} \int_0^1 \frac{dy}{(1-y)^{1/2}} f^*(x\phi^*, yR_{\max}), \quad (67)$$

where n_0 is the density of stars in the cluster core, $\phi^* = GM/r\langle v^2\rangle$, and $R_{\max} = 4x(1-x)$. The density of the unbound Maxwellian population is given by

$$n_u(r) = n_0[2\pi^{-1/2}\phi^{*1/2} + \exp(\phi^*)\,\text{erfc}\,(\phi^{*1/2})], \quad (68)$$

where erfc (x) is the complement of the error function. In the limit $r \ll r_h \equiv GM/\langle v^2\rangle$, the first term in the brackets dominates the second, and $n_u \propto r^{-1/2}$. (For $M = 10^3\,M_\odot$ and $\langle v^2\rangle = 100 \text{ km}^2 \text{ s}^{-2}$, $r_h = 1.3 \times 10^{17}$ cm.)

A measure of the anisotropy of the velocity distribution at a radius r is provided by comparison of the radial and tangential velocity dispersions, denoted respectively by $\langle v_r^2\rangle$ and $\langle v_t^2\rangle$. For the isotropic distribution of unbound stars, $\langle v_r^2\rangle = \frac{1}{2}\langle v_t^2\rangle$. Integral expressions for the velocity dispersions are

$$\langle v_r^2(r)\rangle = 2\pi^{-1/2} v_0^2 \phi^{*5/2} \frac{n_0}{n(r)} \int_0^1 dx(1-x)^{3/2} \int_0^1 dy(1-y)^{1/2} f^*(x\phi^*, yR_{\max}),$$
$$\langle v_t^2(r)\rangle = 2\pi^{-1/2} v_0^2 \phi^{*5/2} \frac{n_0}{n(r)} \int_0^1 dx(1-x)^{3/2} \int_0^1 dy\, y(1-y)^{-1/2} f^*(x\phi^*, yR_{\max}), \quad (69)$$

where v_0^2 denotes the one-dimensional core velocity dispersion.

Figure 7 indicates the run of $n(r)$ computed numerically using equation (67) for the distribution shown in Figure 3. The contribution to $n(r)$ from the unbound stars given by equation (68) has been included. The density increases monotonically with decreasing r until $r \approx 3 \times 10^{-5} r_h$. At $r = r_h$, the density is a factor of 2.6 larger than the core density of the cluster; this enhancement is approximately equal to the Boltzmann factor, $\exp[\phi^*(r_h)] \approx 2.7$. The logarithmic derivative $d \ln n/d \ln r \approx -1.65$ for r in the range $10^{-3} < r/r_h < 10^{-1}$. This is in reasonable agreement with the results of Bahcall and Wolf (1976, 1977), who find that $d \ln n/d \ln r = -1.75$, which is the value predicted by the $f \propto E^{1/4}$ power law distribution function. Shapiro and Marchant (1978) also obtain a value of -1.75 for the logarithmic derivative of the density profile. In contrast, Ipser (1978*a*, *b*) finds that $d \ln n/d \ln r = -1.25$ over most of the range of r.

Since the density profile $n(r)$ is well approximated by a power law over most of the range of r, the surface density $\sigma(r)$ is expected to have a power law profile which is flatter by about one power of r. Numerical integration indicates that our surface density is reasonably approximated by $\sigma \propto r^{-0.5}$.

Figure 8 illustrates the dependences of $\langle v_r^2\rangle$ and $\langle v_t^2\rangle$ on r. The contributions from both bound and unbound stars are included. As expected, the velocity dispersions begin to deviate from isothermal behavior for $r \lesssim r_h$, where the potential of the black hole exceeds the core velocity dispersion. Due to the progressive depletion of stars in small R orbits, with increasing energy, the anisotropy of the velocity distribution grows with decreasing r. For $r < 10^{-4} r_h$, $\langle \frac{1}{2} v_t^2\rangle \geq 2\langle v_r^2\rangle$. For r in the range $10^{-3} < r/r_h < 10^{-1}$, both velocity dispersions are reasonably approximated by $\langle v_r^2\rangle \approx \frac{1}{2}\langle v_t^2\rangle \approx 0.4 v_0^2 r_h/r$.

VII. DISCUSSION

The technique by which we have determined the steady-state star distribution around a black hole—the direct numerical integration of a Fokker-Planck equation in (energy, angular momentum)-space—has not previously been used in stellar dynamics. It is interesting to compare our approach with the two-dimensional Monte Carlo

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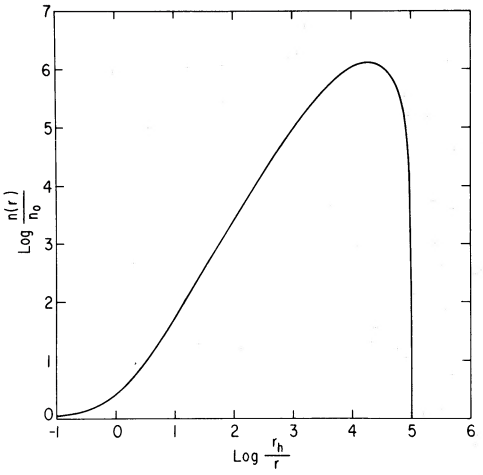


FIG. 7

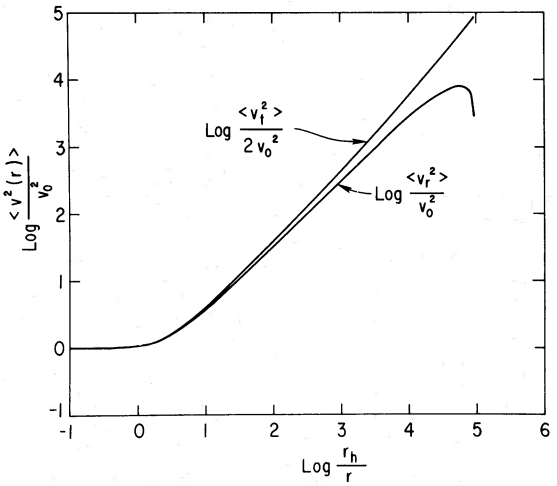


FIG. 8

FIG. 7.—The stellar density profile for the distribution of Fig. 3. The density is normalized by the cluster core density n_0. The radius is normalized by $r_h \equiv GM/\langle v^2 \rangle$. For the chosen cluster parameters, $n_0 = 5 \times 10^4$ stars pc^{-3} and $r_h = 1.3 \times 10^{17}$ cm. The contribution to the density from the unbound Maxwellian cluster core population is included.

FIG. 8.—The velocity dispersion profiles for the distribution of Fig. 3. The radial and tangential velocity dispersions are normalized by the cluster core velocity dispersion $v_0^2 = 100$ km^2 s^{-2}. The contributions from unbound core stars are included. Note the increasing anisotropy with decreasing r.

simulations of the star distribution around a black hole carried out by Shapiro and Marchant (1978). While the results of these two studies are in reasonable (and gratifying) agreement, we feel that direct numerical integration presents several advantages over Monte Carlo simulations for treating this type of problem.

In terms of computer time, direct numerical integration is significantly more economical than Monte Carlo simulations for *two-dimensional* problems. Shapiro and Marchant (1978) report that their code required about 45 minutes of CDC 6600 time per iteration, with two iterations required to produce a self-consistent steady-state solution. Our code requires about 1 minute of IBM 370/158 time per time step; convergence to a steady-state solution occurs in about three time steps even when the initial distribution is quite different from the steady-state solution. Since the two machines used have similar computational speed, the direct numerical integration requires only about 1/30 the time of the Monte Carlo simulation.

Direct numerical integration of the Fokker-Planck equation avoids the statistical uncertainty (and consequent graininess) inherent in Monte Carlo solutions. It is also more straightforward to enforce physical boundary conditions in a finite difference scheme than in a discrete simulation. For example, the vanishing of our R-flux coefficients at the $R = 1$ (circular orbit) boundary naturally enforces the physical condition that there be no flux through it. In contrast, this boundary presents some difficulties for the Monte Carlo scheme, as noted by Shapiro and Marchant (1978).

As discussed by Shapiro and Marchant (1978), their Monte Carlo approach has the potential for being extended both to include the self-potential of the star distribution and to treat large-angle scatterings. Our finite difference scheme can also be extended in these ways. It is fairly straightforward to obtain general expressions for the E and R flux coefficients in an arbitrary spherically symmetric potential. A procedure is presently being developed for simultaneously advancing the distribution function and the potential in time using Poisson's equation in addition to the Fokker-Planck equation. In principle, large angle scatterings may be treated by use of rate coefficients $R_{ij,kl}$ for the transfer of stars from grid cell ij to cell kl. The finite difference equations would take the same form as equation (52) except that the sums would now extend over all cells rather than merely over adjacent cells. The required matrix inversion would, of course, be significantly more time-consuming in this case.

As noted above, the results of Ipser (1978*a*, *b*) are in distinct disagreement with those presented here, with the similar results reported by Shapiro and Marchant (1978), and with the earlier results of Bahcall and Wolf (1977) and Lightman and Shapiro (1977). Ipser attributes his lack of agreement with the studies previous to his own, to approximations made in these studies. He cites the one-dimensional character of the Bahcall and Wolf (1977) analysis and the use by Lightman and Shapiro (1977) of a power law background distribution in the calculation of diffusion coefficients as reasons for the discrepancy. Ipser's criticisms do not apply to the present two-dimensional calculation nor to that of Shapiro and Marchant (1978).

We note that Ipser (1978*a*, *b*) calculates only the angle-averaged part $\bar{f}(r, v)$ of the general distribution function $f(r, v, \mu)$ ($\mu \equiv \hat{\mathbf{r}} \cdot \hat{\mathbf{v}}$); i.e., he develops and solves an equation only for the $P_0(\mu)$ component in a Legendre expansion of f. In doing this, he neglects the Vlasov terms which strongly couple the orders of such an expansion. We feel

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that the fully two-dimensional treatment of the test distribution presented here and that of Shapiro and Marchant (1978) are inherently more reliable than Ipser's approach.

VIII. SUMMARY

In this paper we have presented a determination of the steady-state stellar distribution around a black hole by application of the orbit-averaged Fokker-Planck equation. We obtained this equation from the full Boltzmann equation by use of an expansion of the distribution function in the ratio of orbital period to relaxation time. The lowest order of the expanded Boltzmann equation was shown to be satisfied by any function f_0 of energy and angular momentum. By considering the next order of the Boltzmann equation we obtained a kinetic equation for the slow time evolution of f_0. This equation was shown to be identical to the orbit-averaged Fokker-Planck equation. If the angular momentum dependence of f_0 is neglected and the orbit-averaged equation is further averaged over angular momentum, we obtain the one-dimensional kinetic equation for $f_0(E, t)$ treated by Bahcall and Wolf (1976).

In our numerical treatment of the Fokker-Planck equation we made one basic approximation: we calculated the collision integrals using a fictitious isotropic background distribution set equal to the angular momentum average of the actual distribution, as did Shapiro and Marchant (1978). The expansion of the distribution function is inappropriate near the loss cone region of phase space, so we developed a boundary layer analysis from which we obtained a boundary condition for the solution of the orbit-averaged Fokker-Planck equation in the main region. Although the non-Keplerian nature of the potential at large radii was not rigorously taken into account, we argued that the solution of our Fokker-Planck equation should approach the Maxwellian core distribution in the small energy limit. This assertion was substantiated by our numerical results.

The results of our numerical integrations of the Fokker-Planck equation are: steady-state distribution functions (Figs. 3–6); the stellar density profile (Fig. 7); the radial and tangential velocity dispersions (Fig. 8); and rates for the loss of stars and energy to the black hole, and for physical collisions (Table 1). A simplified analysis of the diffusion of stars into the loss cone led to a formula giving the dependence of the loss rate on cluster parameters (eq. [66]).

Our two-dimensional steady-state distribution is in reasonable agreement with the isotropic single mass distribution of Bahcall and Wolf (1977) at low energies ($E \lesssim 100\langle v^2 \rangle$) but begins to show significant anisotropy and depletion below the Bahcall and Wolf distribution at larger energies. The angular momentum averaged energy profile of our distribution agrees reasonably well with that obtained by Shapiro and Marchant (1978) from their Monte Carlo simulations of the star distribution.

The rate of tidal destruction of stars which we obtained for a $10^3\ M_\odot$ black hole in a "standard" X-ray globular cluster—30 stars per 10^8 yr—is about a factor of 2 larger than that found by Shapiro and Marchant (1978). As discussed by Shapiro (1977), a tidal destruction rate of this magnitude has significant consequences for the evolution of the cluster. We find that the physical collision rate among the stars bound to the black hole is smaller than the tidal destruction rate by a factor of 30 for a $10^3\ M_\odot$ black hole in a standard cluster.

The profile of the density cusp surrounding the black hole was found to be reasonably approximated by a power law, $n \propto r^{-1.65}$, which is slightly flatter than the $r^{-7/4}$ power law found by Bahcall and Wolf (1976). Thus their conclusions regarding the possibilities for detecting black holes in globular clusters apply to our results.

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APPENDIX

THE ORBIT-AVERAGED FOKKER-PLANCK EQUATION

In a general velocity-space coordinate system (v^i), the single-species Fokker-Planck equation takes the form (Rosenbluth, MacDonald, and Judd 1957)

$$\left(\frac{\partial f}{\partial t}\right)_c = -(f\langle \Delta v^i \rangle)_{;i} + \tfrac{1}{2}(f\langle \Delta v^i \Delta v^j \rangle)_{;ij}\,, \tag{A1}$$

where the semicolons indicate covariant differentiation, the summation convention is assumed, and the velocity diffusion coefficients are given by

$$\langle \Delta v^i \rangle = \Gamma h^{,i}\,, \qquad \langle \Delta v^i \Delta v^j \rangle = \Gamma g^{,ij}\,, \tag{A2}$$

with

$$g(\boldsymbol{v}) = \int d^3v' f(\boldsymbol{v}')|\boldsymbol{v} - \boldsymbol{v}'|\,, \qquad h(\boldsymbol{v}) = 2\int d^3v' f(\boldsymbol{v}')|\boldsymbol{v} - \boldsymbol{v}'|^{-1}\,. \tag{A3}$$

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The Fokker-Planck equation can equivalently be written

$$\left(\frac{\partial f}{\partial t}\right)_c = -a^{-1/2}[a^{1/2}f(\langle\Delta v^i\rangle - \tfrac{1}{2}\Gamma^i{}_{jk}\langle\Delta v^j\Delta v^k\rangle)]_{,i} + \tfrac{1}{2}a^{-1/2}[a^{1/2}f\langle\Delta v^i\Delta v^j\rangle]_{,ij} \tag{A4}$$

where the commas indicate partial differentiation, $a = \det(a_{ij})$, a_{ij} is the velocity space metric, and $\Gamma^i{}_{jk}$ is the corresponding connection coefficient.

The variables E and R introduced in § II*a* may be regarded as velocity-space coordinates at a given spatial point $\boldsymbol{r}$. Then the determinant of the metric tensor is given by

$$a^{1/2} = \frac{\pi(GM)^2}{r^2 v_r E}\,. \tag{A5}$$

We recall that the first term of the λ expansion of f introduced in § I*a* can be written $f_0 = f_0(E, R, t)$. In (E, R) coordinates, the local (r-dependent) Fokker-Planck equation takes the form

$$a^{1/2}\left(\frac{\partial f_0}{\partial t}\right)_c = -\frac{\partial}{\partial E}(a^{1/2}f_0\langle\Delta E\rangle) + \frac{1}{2}\frac{\partial^2}{\partial E^2}(a^{1/2}f_0\langle(\Delta E)^2\rangle) - \frac{\partial}{\partial R}(a^{1/2}f_0\langle\Delta R\rangle) + \frac{1}{2}\frac{\partial^2}{\partial R^2}(a^{1/2}f_0\langle(\Delta R)^2\rangle) + \frac{\partial^2}{\partial E\partial R}(a^{1/2}f_0\langle\Delta E\Delta R\rangle)\,, \tag{A6}$$

where

$$\begin{aligned}
\langle\Delta E\rangle &= E_{,i}\langle\Delta v^i\rangle + \tfrac{1}{2}E_{;ij}\langle\Delta v^i\Delta v^j\rangle\,,\\
\langle(\Delta E)^2\rangle &= E_{,i}E_{,j}\langle\Delta v^i\Delta v^j\rangle\,,\\
\langle\Delta R\rangle &= R_{,i}\langle\Delta v^i\rangle + \tfrac{1}{2}R_{;ij}\langle\Delta v^i\Delta v^j\rangle\,,\\
\langle(\Delta R)^2\rangle &= R_{,i}R_{,j}\langle\Delta v^i\Delta v^j\rangle\,,\\
\langle\Delta E\Delta R\rangle &= E_{,i}R_{,j}\langle\Delta v^i\Delta v^j\rangle\,.
\end{aligned} \tag{A7}$$

For definiteness, the (v^i) may be taken to represent the (v, μ, ϕ) coordinate system for which the velocity diffusion coefficients are calculated by Rosenbluth, MacDonald, and Judd (1957).

The orbit-averaged Fokker-Planck equation is obtained by integrating the local equation (A6) over the spatial volume filled by star orbits:

$$\int_{r_-}^{r_+} dr 4\pi r^2 a^{1/2}\left(\frac{\partial f_0}{\partial t}\right)_c = -\int_{r_-}^{r_+} dr 4\pi r^2 \frac{\partial}{\partial E}(a^{1/2}f_0\langle\Delta E\rangle) + \cdots\,. \tag{A8}$$

It is only necessary to consider one term on the right-hand side of equation (A8) since each term is treated in the same fashion. Recalling the definition of the orbital integral (eq. [8]), (A8) may be written

$$2\pi^2(GM)^2E^{-1}\oint\frac{dr}{v_r}\left(\frac{\partial f_0}{\partial t}\right)_c = -\oint dr\,2\pi r^2\frac{\partial}{\partial E}(a^{1/2}f_0\langle\Delta E\rangle) + \cdots\,. \tag{A9}$$

Since the integrals are cyclic, the r integration may be interchanged with the energy differentiation without introducing any endpoint terms. Thus,

$$2\pi^2(GM)^2E^{-1}\oint\frac{dr}{v_r}\left(\frac{\partial f_0}{\partial t}\right)_c = -\frac{\partial}{\partial E}\oint\frac{dr}{v_r}2\pi^2(GM)^2E^{-1}f_0\langle\Delta E\rangle + \cdots\,. \tag{A10}$$

We recall that the period of a Keplerian orbit is $P = 2^{-1/2}\pi GME^{-3/2}$. Thus (A10) may be written

$$2^{1/2}\pi^3(GM)^3E^{-5/2}P^{-1}\oint\frac{dr}{v_r}\left(\frac{\partial f_0}{\partial t}\right)_c = -\frac{\partial}{\partial E}\left[2^{1/2}\pi^3(GM)^3E^{-5/2}f_0P^{-1}\oint\frac{dr}{v_r}\langle\Delta E\rangle\right] + \cdots\,. \tag{A11}$$

Let

$$\left(\frac{\partial N}{\partial t}\right)_c \equiv 2^{1/2}\pi^3(GM)^3E^{-5/2}P^{-1}\oint\frac{dr}{v_r}\left(\frac{\partial f_0}{\partial t}\right)_c\,, \tag{A12}$$

where $N(E, R) = 2^{1/2}\pi^3(GM)^3E^{-5/2}f_0(E, R)$ is the number density in (E, R)-space, and let $(\partial N/\partial t)_p$ be similarly

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defined. Also let

$$\langle \Delta E \rangle_t \equiv P^{-1} \oint \frac{dr}{v_r} \langle \Delta E \rangle \,. \tag{A13}$$

Then the nonlocal Fokker-Planck equation (9) takes the form

$$\begin{aligned} \frac{\partial N}{\partial t} &= \left(\frac{\partial N}{\partial t}\right)_c + \left(\frac{\partial N}{\partial t}\right)_p \\ &= -\frac{\partial}{\partial E}\left(N\langle \Delta E \rangle_t\right) + \frac{1}{2}\frac{\partial^2}{\partial E^2}\left(N\langle (\Delta E)^2 \rangle_t\right) - \frac{\partial}{\partial R}\left(N\langle \Delta R \rangle_t\right) \\ &\quad + \frac{1}{2}\frac{\partial^2}{\partial R^2}\left(N\langle (\Delta R)^2 \rangle_t\right) + \frac{\partial^2}{\partial E \partial R}\left(N\langle \Delta E \Delta R \rangle_t\right) + \left(\frac{\partial N}{\partial t}\right)_p . \end{aligned} \tag{A14}$$

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